

Avr-Gcc Assignment

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I. ABSTRACT

This paper explains about the question tests the validity of Boolean identities involving the XOR ($\oplus$) operator. Four expressions are given, and the objective is to identify which one does not represent a valid identity. The provided options explore properties such as associativity, distributivity, and specific conditions for XOR operations.

- 1) $(x \oplus y) \oplus z = x \oplus (y \oplus z)$
- 2) $(x + y) \oplus z = x \oplus (y + z)$
- 3) $x \oplus y = x + y, \text{ if } xy = 0$
- 4) $x \oplus y = (xy + x'y)'$

The question can be implemented using avr-gcc with arduino uno and lcd.

II. COMPONENTS

The required components list is given in Table: I., pin diagram of Led is shown in Fig.1.

Components	Value	Quantity
Arduino	UNO	1
lcd		1
Jumper Wires		20
Breadboard		1

TABLE I

III. PROCEDURE

- 1) Make connections of arduino uno to led as shown in below fig-2.

Arduino UNO	LCD
pin-13	Anode
gnd	cathode

TABLE II

- 2) pin configuration of lcd.
- 3) Give two inputs called x and y separately in bread board and make short connections of inputs and vcc and gnd.
- 4) By providing proper inputs observe the blinking of led for output 1 and 0 as per below truth table of xor operation.
- 5) Execute the avr-gcc code in nvim editor using make command.
- 6) After upload the code into hardware setup using arduino IDE platform with .hex file.

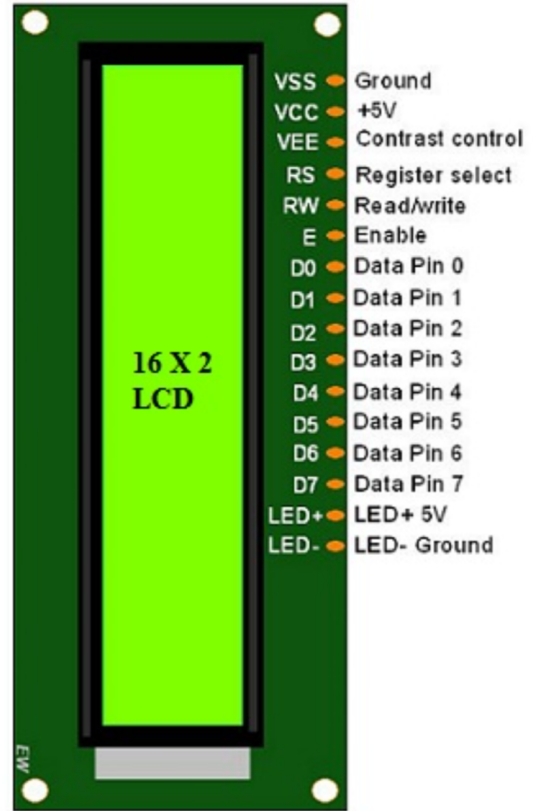


Fig. 1.

x	y	$x \oplus y$
0	0	0
0	1	1
1	0	1
1	1	0

TABLE III

IV. RESULTS

- 1) Download the codes given in the link below and execute them to see the output as shown in figure 2.
- 2) <https://github.com/madhureddy727/FWC-Assignments>

V. CONCLUSION

Hence implementation of avr-gcc code in arduino uno and the verification of xor truth table is done using led.

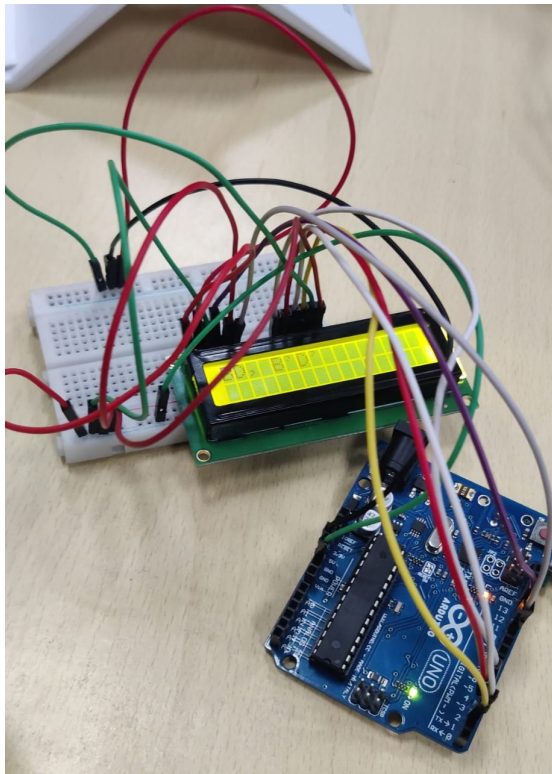


Fig. 2.